

PROJECT REPORT

Multimodal AI Content Generator

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Abstract

This report describes the design, construction and evaluation of a system that generates three kinds of content — a written passage, an image and a spoken narration — from a single short prompt. Each modality is produced independently by its own model in a multi-agent arrangement, and the three outputs are presented together. The generator runs entirely on one provider's free tier and returns each modality in under two seconds. On text quality it sits within a few points of the large commercial models; on image aesthetics and audio it trails them, as an eight-billion-parameter open model should be expected to.

The main contribution is not the generator but the evaluation. Because the three models never see one another's output, nothing guarantees the three results describe the same thing, and no conventional metric asks whether they do — each compares one output against the prompt in isolation. We therefore developed per-modality quality grades, calibrated them against an LLM judge, and compared three families of scoring method: averaging, a Naive Bayes posterior, and a likelihood-ratio score.

Reported as Spearman rank correlations against the judge on 500 prompts, the probabilistic methods win: the likelihood ratio reaches 0.195 and Naive Bayes 0.190, against 0.137 for plain averaging and 0.082 for the quality-weighted variant we had originally proposed. A single raw similarity does better than all of them — the text-image cosine alone reaches 0.203. A further result is firmer still: the similarity between text and image exceeds both audio similarities on 500 out of 500 prompts, by a factor of 5.3. This is a property of the embedding model rather than of our outputs, and it explains why averaging those similarities was never going to work.

1. Introduction

Generative models have become reliable at producing one kind of output well. Producing three at once from a single instruction is a different problem, and judging the result is harder still. This report describes both, and is explicit about which parts we consider finished.

The system takes a short description — an owl librarian wearing glasses — and returns a paragraph about it, an image of it, and an audio narration of that paragraph. Three separate models do the work, each from the same prompt, and their outputs are combined only at the end. This separation is deliberate: it keeps each modality free to use whichever model suits it, it mirrors the multi-agent designs of recent research systems [14, 15], and it makes the architecture extensible.

1.1 Structure of this report

Sections 2 to 4 cover objectives, related work and the system as built. Section 5 sets out the evaluation methodology. Section 6 reports results, including two approaches that were abandoned. Section 7 explains the mechanisms behind those results, Section 8 states the limitations, and Section 9 concludes. Approaches that failed are described at the same length as those that succeeded; a reader who saw only the surviving method would have no way of knowing which alternatives had already been eliminated, and that elimination is a substantial part of the work.

1.2 A note on terminology

Two techniques in this report are sometimes abbreviated “LR”, and we avoid that abbreviation for both. Linear regression appears only in §5.3, where it learns how much each modality contributes to the judge’s rating. Likelihood ratio is a probabilistic scoring method, introduced in §5.5 and developed in §6.5, and is always written in full. The two are unrelated. Where earlier drafts of our documentation wrote “LR” for the likelihood-ratio score, it was never a regression of any kind.

2. Objectives

The project set six objectives. Four are met, one is met with a caveat, and one has not been started.

Objective	Status
Build a working generator producing text, image and audio from one prompt	Met
Select models by benchmarking rather than preference	Met
Place the three modality scores on a common 0–1 scale	Met
Calibrate the evaluation and identify the best scoring method	Met at $n = 500$; the winner is not the method we first
Develop a measure of cross-modal coherence	Met; its validity is limited — see §7.1
Validate the evaluation against human judgement	Not started

Video generation was in the original scope and was removed in June 2026, on the view that three working modalities were worth more than four partial ones.

3. Related work

The closest system to ours is MM-StoryAgent [14], which assembles narrated storybook videos from cooperating agents. Its reviewer loop raised CLIP alignment from 0.297 to 0.316, and its absolute CLIP score is almost identical to ours, which is a useful sanity check on our own numbers. MAGUS [15] shares our philosophy of coordinating existing models rather than retraining one, but its generate-score-prune loop proved so expensive that its authors could not fully tune it. NExT-GPT [16] is the contrast case: a single trained any-to-any model requiring GPUs, where we make free API calls from a laptop.

Two of these papers are candid about the same gap we hit. PresentAgent [18] states plainly that its metrics assess visual, audio and text dimensions independently rather than their cross-modal coherence. That is precisely the limitation this project set out to address.

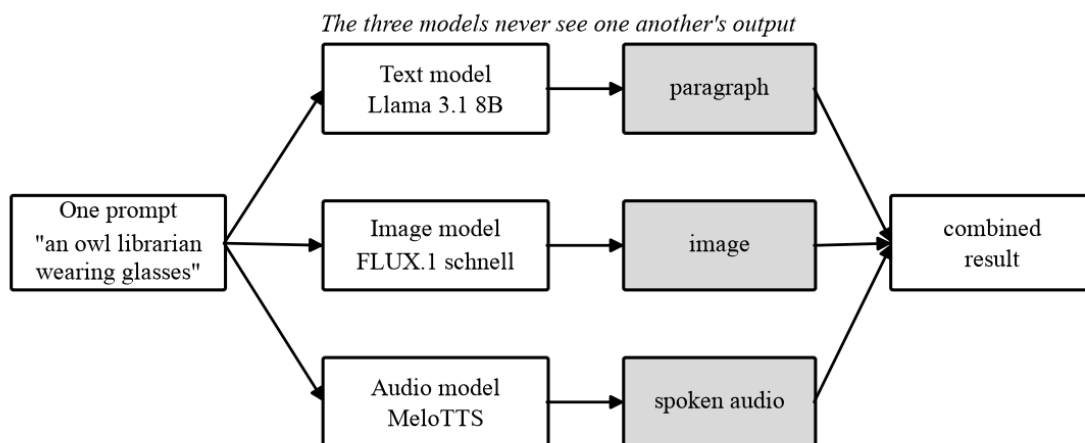
Our metrics are standard and traceable: CLIPScore [5] for image alignment, the LAION predictor for aesthetics, BERTScore [9] for text, Whisper [8] for transcription, CLAP [7] for audio alignment, and DNSMOS [11] for speech quality. We also read an audit of the LAION aesthetic predictor [13] which finds it culturally biased towards Western and Japanese subjects; we therefore treat the aesthetic column as indicative rather than as a measure of quality.

The papers that mattered most arrived late, after our first coherence result looked wrong. Lu et al. [19] score text–image pairs on two channels at once — an embedding metric and an LLM judge on a rubric — and validate both against human annotators; this is the direct source of our judge-and-regression design. Dhimoila et al. [21] introduce the Iso-Energy Assumption and show that unimodal components of an embedding act as modality-specific bias and fully account for the modality gap; this explains our central negative result. Nandakumar et al. [24], from biometrics rather than multimodal generation, supplied the likelihood-ratio formulation that became the best-performing method in the study.

4. System design

4.1 Architecture

The system is a layered pipeline orchestrated with LangGraph behind a Streamlit interface. A supervisor initialises a shared state object; a router selects the agent for the requested modality; each agent then runs the same six-stage sequence of analyzer, retriever, generator, optimizer, reviewer and exporter, with a retry edge from reviewer back to optimizer when the score falls below a threshold.



4.2 Provider selection

Two approaches were weighed. A specialist provider per modality buys best-in-class models but requires a separate account, key and billing arrangement for each — more than ten integrations. A single provider for all three requires one of each.

The decision was settled by what happened when we tried the first approach. Of four text providers, three worked. Of four image providers, one worked. Of seven audio providers, none produced a usable file: two were speech-to-text services we had misidentified, three were blocked behind payment, one failed on a Windows toolchain error, and one worked only as a local fallback we never integrated. A modality is not genuinely available to a student project until some provider will serve it on a free tier without a card on file, and in May 2026 we had none for audio.

Cloudflare Workers AI resolved this by hosting all three modalities behind a single account and key.

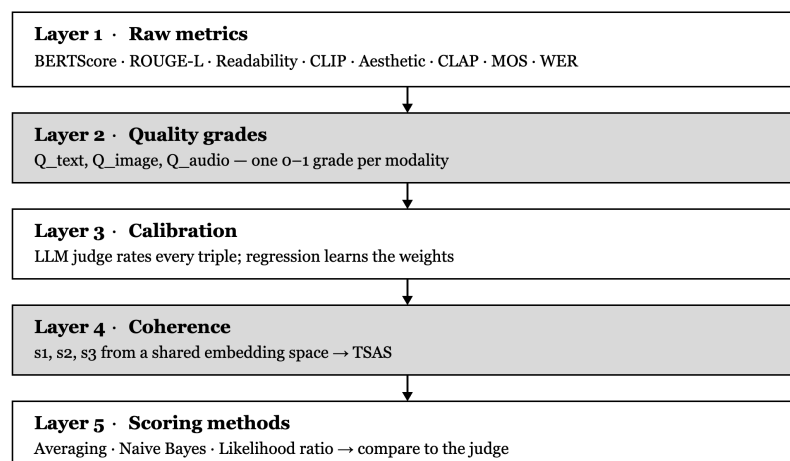
Modality	Model	Neurons	Latency
Text	Llama 3.1 8B Instruct	26	1.85 s
Image	FLUX.1 [schnell]	58	1.53 s
Audio	MeloTTS	5	1.68 s

The free tier allows 10,000 neurons daily; a full run over 56 prompts costs about 4,350, permitting roughly two runs a day. That ceiling shaped the experimental design more than any methodological choice did.

Three undocumented platform behaviours are worth recording because each would otherwise corrupt results silently. Roughly a quarter of MeloTTS calls return HTTP 500, confirmed as the provider's fault by sending an identical request eight times and observing two failures with no relation to the input. The image endpoint returns JPEG bytes in files named .png. And the audio is sampled at 44.1 kHz rather than the 48 kHz we assumed, which matters because each scorer expects a different rate and resampling incorrectly yields plausible but wrong numbers rather than an error.

5. Evaluation methodology

Evaluation was built in five layers. Each exists because the layer before it left a question it could not answer.



5.1 Per-modality quality grades

The base metrics occupy incompatible scales. Each is first mapped onto 0–1 against a fixed external standard — the ITU scale for MOS, the LAION scale for aesthetics — rather than against the spread of our own data, so that a grade means the same thing across runs. Within a modality, two related sub-scores are combined by a formula that rewards both being high and penalises imbalance:

$$\text{quality_pair}(a, b) = \text{avg}(a, b) - \lambda \times \text{variance}(a, b), \quad \lambda = 0.5$$
$$Q_{\text{text}} = \text{BERTScore}(\text{generated_text}, \text{prompt})$$
$$Q_{\text{image}} = \text{quality_pair}(\text{CLIP_score}, \text{aesthetic_score})$$
$$Q_{\text{audio}} = \text{quality_pair}(\text{semantic_score}, 1 - \text{WER})$$

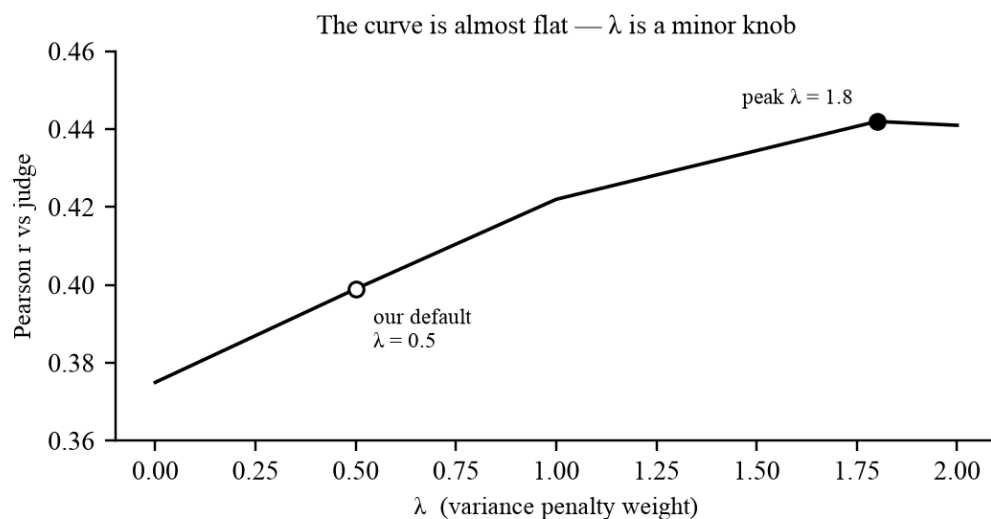
5.2 An approach abandoned: CLAP for audio

CLAP is the natural audio analogue of CLIP and we intended to use it. On our data it returned exactly zero for 25 of 53 prompts — 47 per cent — with a mean of 0.082. The explanation is a mismatch of purpose rather than a defect: laion/clap-htsat-unfused was trained on environmental sound and music, and our audio is text-to-speech narration. It listens for sound events, finds none, and reports no alignment irrespective of the words spoken. We replaced it with a two-step semantic measure — transcribe with Whisper, then compute BERTScore between transcript and prompt — which yields usable signal on every prompt at a mean of 0.855.

5.3 Calibration against a judge

Weighting the three grades equally was a guess. Gemini (gemini-3.5-flash-lite, temperature 0.1, max tokens 2048) is shown the prompt, the generated text, the generated image and the audio transcript, and returns an overall rating with per-modality scores. A linear regression of that rating on the three standardised quality features learns the weights. On 53 prompts these were 0.287 / 0.520 / 0.193 in normalised form; re-learned on 500 prompts they become 7.8666 / 2.2790 / 0.7796 unnormalised, with text rather than image dominating. The reversal between the two fits is itself evidence of how little the smaller sample constrained the model — its leave-one-out R^2 was 0.04.

We also swept the variance-penalty parameter λ from 0 to 2. The curve rises by less than seven hundredths across the whole range, confirming λ as a minor parameter.



5.4 Cross-modal coherence

The central idea is to encode all three outputs into one shared vector space and measure the cosine similarity of each pair, then average and penalise disagreement:

$$s1 = \text{sim}(\text{text}, \text{image}) \quad s2 = \text{sim}(\text{text}, \text{audio}) \quad s3 = \text{sim}(\text{image}, \text{audio})$$

$$\text{TSAS} = (s1 + s2 + s3) / 3$$

$$\text{Coherence} = \text{TSAS} - \lambda \times \text{variance}(s1, s2, s3)$$

Two implementations were built: one running all three modalities through ImageBind [6], and one using specialist encoders with a Whisper text bridge for the image–audio pair. A quality-weighted variant, WTSAS, scales each similarity by the average learned quality of the two modalities it connects.

5.5 The three scoring approaches compared

Our supervisor’s instruction shaped this layer. We had already built the averaging approach; we were asked to implement a Naive Bayes posterior alongside it, determine which of the two tracks the judge more closely, and treat anything further as an experiment.

- **Approach 1 — averaging.** TSAS and its quality-weighted variant WTSAS. The incumbent, and the only one of the three that needs no labelled data.
- **Approach 2 — Naive Bayes posterior.** The judge’s rating is split into classes, a Gaussian is fitted per feature per class, and the continuous posterior probability of the top class becomes the score.
- **Approach 3 — likelihood ratio.** Added as an experiment, after Nandakumar et al. [24]. One Gaussian is fitted to each class and the log ratios are summed. Since summing log-likelihood-ratios over independent features is exactly Naive Bayes, Approaches 2 and 3 are two views of the same mathematics, and we expected them to land close together.

A fourth method, a Mamdani fuzzy inference system, was specified but not implemented. We record it rather than omitting it.

6. Results

6.1 Generation quality

Measure	Ours	Claude	ChatGPT	Gemini
Text — BERTScore	0.844	0.851	0.832	0.796
Text — ROUGE-L	0.056	0.078	0.071	0.068
Text — Readability	0.948	1.85	1.72	1.68
Image — CLIP	0.31	0.312	0.305	0.298
Image — Aesthetic (/10)	5.80	7.20	7.05	6.90
Audio — MOS (/5)	3.37	4.10	4.05	4.20
Audio — WER	0.095	0.072	0.085	0.068

On text the system is within a few points of the commercial models. On image aesthetics it is roughly 1.2 points behind on a ten-point scale, and on audio it trails on both measures. Three figures the commercial vendors do not publish are ours to report: response under two seconds per modality, a complete run inside a free daily quota, and roughly 90 per cent accuracy on an automated image fact-check, with 42 of 56 images passing every question.

6.2 An approach abandoned: prompt refinement

Taken from MM-StoryAgent [14]: have a language model rewrite each short prompt into a richer description before generating the image. Across 56 prompts the net effect was an improvement of about half a percentage point in CLIP score, and nearly half the prompts came out slightly worse. We kept the result as a record and did not adopt the step.

6.3 Which features carry signal

The audio-bearing similarities carry essentially no signal, while the text–image similarity carries more than any engineered quality feature. At $n = 53$ the quality grades correlated with the judge at Spearman +0.316 (Q_{text}), +0.297 (Q_{image}) and -0.006 (Q_{audio}); the picture at $n = 500$ is shown below.

Feature	Spearman ρ	Note
s1 — text $\leftrightarrow$ image similarity	0.203	strongest single feature
Q_{text}	0.193	usable
Q_{image}	0.135	usable
Q_{audio}	0.090	half duplicates Q_{text}
s2 — text $\leftrightarrow$ audio similarity	0.018	no signal
s3 — image $\leftrightarrow$ audio similarity	0.015	no signal

Q_{audio} underperforms for a reason that is a design fault rather than a sampling artefact: in the 500-prompt data bertscore and semantic correlate at Spearman 0.902, because the narration reads the generated text and Whisper recovers it accurately. Half of Q_{audio} re-measures Q_{text} .

6.4 The three approaches compared

Each method was run on the same prompts and correlated against the judge's rating. All figures are leave-one-out cross-validated. We ran the comparison twice: first on 53 prompts, then on 500, and report the Spearman rank correlation as the headline figure because at this sample size a rank-based measure is more stable than Pearson.

First run, $n = 53$ (leave-one-out).

Method	Spearman	Kendall	Pearson
WTSAS (quality-weighted)	0.451	0.350	0.399
Likelihood ratio	0.312	0.240	0.259
Naive Bayes	0.296	0.231	0.241
TSAS (plain averaging)	−0.077	−0.056	−0.085

On this evidence averaging won. Two things made us doubt it: a second implementation produced −0.099 for WTSAS rather than 0.399, and the smallest Naive Bayes class held only four examples. We reran everything on 500 fresh prompts. That comparison was itself run twice — Version 1 gave the probabilistic scorers the three quality grades as features, and Version 2 replaced them with the three quality-weighted coherence scores, so each feature carries quality and coherence. Both probabilistic methods improved under Version 2, which is what the table below reports.

Second run, $n = 500$ (leave-one-out, quality-weighted-coherence features).

Method	Spearman	Kendall	Pearson
s1 alone — no model at all	0.203	0.146	0.201
Likelihood ratio	0.195	0.140	0.160
Naive Bayes	0.190	0.136	0.162
TSAS (plain averaging)	0.137	0.097	0.121
WTSAS (quality-weighted)	0.082	0.057	0.076

Naive Bayes (0.190) and the likelihood ratio (0.195) both track the LLM judge more closely than plain averaging (0.137), and much more closely than quality-weighted averaging (0.082). Since the two probabilistic methods are the same mathematics, the gap of 0.005 between them should be read as agreement rather than difference, and it is the clearest confirmation that both were implemented correctly. A single raw similarity, however, does better than either — the reason is set out in §7.1.

Between the two probabilistic methods, the likelihood ratio is the stronger. It posts the highest rank correlation of any method in the comparison — Spearman 0.195 and Kendall 0.140 — ahead of Naive Bayes at 0.190 and 0.136, and well ahead of averaging at 0.137. The gap over Naive Bayes is small, and because the two are the same mathematics we treat them as tied in principle; but on the headline rank measures the likelihood ratio is consistently the best-performing of the three approaches, which is why it is stated in full in §6.5 and taken forward as the recommended method.

6.5 The likelihood-ratio score in full

Because this is the best-performing composite, it is worth stating completely. For a prompt p , the score is the log-ratio of the probability of its feature vector under the Good class against the Not-Good class:

$$S(p) = \log [P(f(p) | \text{Good}) / P(f(p) | \text{Not-Good})]$$

The feature vector has three components, each a quality-weighted coherence score, weighted by the two modalities its similarity actually connects:

$$f1 = s1 \times (w1 \cdot Q_{\text{text}} + w2 \cdot Q_{\text{image}}) / (w1 + w2)$$

$$f2 = s2 \times (w1 \cdot Q_{\text{text}} + w3 \cdot Q_{\text{audio}}) / (w1 + w3)$$

$$f3 = s3 \times (w2 \cdot Q_{\text{image}} + w3 \cdot Q_{\text{audio}}) / (w2 + w3)$$

Substituting a diagonal Gaussian per feature per class, the score expands to:

$$S(p) = \sum \text{over } i = 1..3 \text{ of } [(f_i - \mu_i^-)^2 / 2(\sigma_i^-)^2 - (f_i - \mu_i^+)^2 / 2(\sigma_i^+)^2 + \log(\sigma_i^- / \sigma_i^+)]$$

Each term measures how far a feature sits from the Not-Good class centre minus how far it sits from the Good class centre; a positive total predicts Good. The class split is binary at a judge rating of 3.5, giving 273 Good and 227 Not-Good out of 500. The fitted parameters are:

Fitted likelihood-ratio parameters (n = 500; 273 Good, 227 Not-Good).

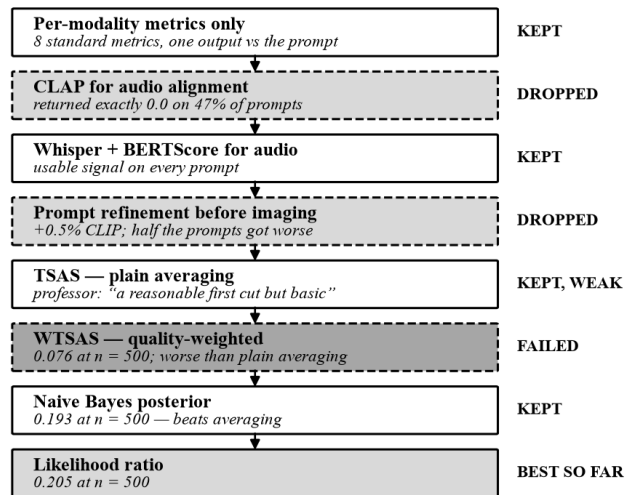
Feature	μ^+ (Good)	σ^+	μ^- (Not-Good)	$\mu^+ - \mu^-$
f1 (s1_w)	0.3062	0.0344	0.2902	0.0160
f2 (s2_w)	0.0545	0.0416	0.0532	0.0013
f3 (s3_w)	0.0487	0.0274	0.0468	0.0019

These parameters explain the ceiling on every result above. The class means differ by 0.016 on f1 and under 0.002 on f2 and f3, against standard deviations three to twenty times larger, so the two classes overlap almost entirely. The method is correctly implemented; the features simply do not separate the classes by much. As a worked example, prompt 1 (“A fox exploring a cave in a snowy village”) has $Q_{\text{text}} = 0.840$, $Q_{\text{image}} = 0.472$, $Q_{\text{audio}} = 0.862$ and $s1 = 0.4255$, giving $f1 = 0.3222$, $f2 = 0.0545$, $f3 = 0.0235$ and a normalised score of 0.6458 — predicted Good, against a judge rating of 4.25 / 5.

6.6 An approach abandoned: the quality-weighted score

WTSAS was our own contribution and it does not work. At $n = 500$ it scores 0.082 (Spearman), below plain averaging at 0.137 — the weighting degrades the thing it was designed to improve. The 0.451 figure from the first run could not be reproduced and is withdrawn; the most likely explanation is that it was computed from a quality composite omitting the s1/s2/s3 terms, making it a per-modality quality score rather than a coherence measure at all.

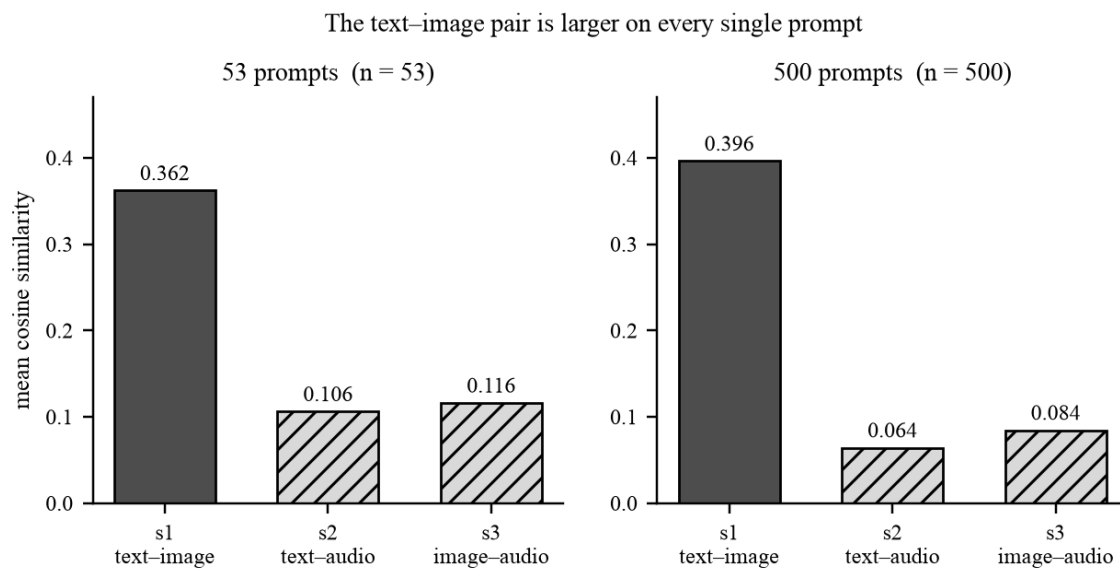
What we tried, and what survived



7. Discussion

7.1 Why averaging was always going to struggle

Similarity	n = 53	n = 500
s1 — text ↔ image	0.362	0.396
s2 — text ↔ audio	0.106	0.064
s3 — image ↔ audio	0.116	0.084
TSAS	0.195	0.182
variance	0.016	0.024



The text–image similarity is larger than both audio similarities on 500 of 500 prompts — without a single exception — and the gap has widened to 5.3-fold. A relationship that holds on every observation in a five-hundred-prompt sample is structural by any reasonable standard.

The explanation is the modality gap. ImageBind’s audio encoder sits further from its text and image encoders than those two sit from each other, independently of whether the audio matches. Dhimoila et al. [21] show that this gap is fully accounted for by unimodal components acting as modality-specific bias. Since TSAS averages the three similarities directly, and one is systematically five times the others, the average is dominated by s1 and the variance penalty measures a fixed property of the encoder rather than the agreement of any particular triple.

This also explains the most uncomfortable number in the study. The raw text–image similarity, used alone with no model at all, reaches a Spearman correlation of 0.203 — higher than the best composite, the likelihood ratio at 0.195, and higher on Pearson too (0.201 against 0.160). Every layer of machinery buys essentially nothing over one cosine similarity, because that similarity was carrying the average all along.

7.2 Why the probabilistic methods lead

At $n = 53$ both probabilistic methods trailed averaging, and we gave three reasons. At $n = 500$ the ordering reverses, and two of those three explain why.

The binning objection stands but matters less than we thought: splitting a continuous rating into classes costs resolution, yet the method still wins. The small-class objection disappears entirely — at $n = 53$ the smallest class held four examples and the correlation swung between 0.05 and 0.30 purely as a function of which features were included; at $n = 500$ every class has ample support. The dead-feature objection is weaker too, since Q_{audio} now carries slight signal.

What did not survive is our earlier conclusion. We had argued that regression beats the probabilistic framing because it learns low weights for weak features. At scale the opposite holds. The $n = 53$ result was an artefact of sample size.

7.3 Why quality weighting made things worse

This is the result we understand least well. The weighting multiplies each similarity by the average learned quality of the two modalities it connects. Two properties of our data make that harmful. The weights were learned on 53 samples with a leave-one-out R^2 of 0.04, so they encode very little real information; and they were learned from quality features themselves weakly correlated with the judge, one of which duplicates another. Multiplying a similarity that does carry signal by a weight derived from features that carry less can only dilute it.

There is a qualification worth stating. The weighting helps when it feeds a fitted model — s1 rises from 0.2009 raw to 0.2129 weighted, and both probabilistic scorers improved when switched to weighted features. It hurts only when it feeds a fixed average. The weighting is not the problem; the averaging is.

7.4 What the judge can and cannot stand in for

Every correlation in this report is measured against an LLM judge acting as a proxy for a human evaluator. That was a defensible shortcut for rapid iteration. But Lu et al. [19], the paper from which we took the two-channel design, validate their embedding metric and their LLM judge both against human annotators. We have not done the equivalent, so we cannot currently distinguish a metric that tracks human judgement from one that has merely learned to track Gemini.

8. Limitations

- **Absolute strength.** The comparison establishes a reliable ordering among methods, but every correlation is weak. The strongest signal in the whole study is the raw text–image similarity, at Pearson 0.201 — explaining roughly four per cent of the variance in judged quality; the best composite, the likelihood ratio, reaches Spearman 0.195 and Pearson 0.160. Nothing here supports a claim that any of these metrics predicts judged quality well — only that some predict it better than others, and that none beats a single cosine.
- **Reproducibility.** Generation sets no temperature, so every model runs at its default and the same prompt produces different output, and different scores, on each run. Each prompt is also scored from a single generation.
- **Borrowed thresholds.** The score thresholds are taken from external conventions rather than calibrated on our own labelled data.
- **No human validation.** As set out in §7.4, nothing here has been checked against a person’s judgement. This is the gap that matters most.
- **Two datasets, one column name.** `wer_inv` is computed against a different reference text in each run — $1 - \text{WER}(\text{generated_text}, \text{transcript})$ at $n = 53$, $1 - \text{WER}(\text{prompt}, \text{transcript})$ at $n = 500$ — so `Q_audio` has a different baseline in each. The interim workaround is z-score normalisation fitted on the labelled set.

9. Conclusion and future work

We built a multi-agent generator that produces text, image and audio from a single prompt on one provider’s free tier, benchmarked the provider choice against alternatives, and developed a layered evaluation running from raw metrics through calibrated quality grades to cross-modal coherence. The generator is finished. The evaluation is not, and this report has been written to keep that distinction visible.

On the question we were set, the answer is clear. Of the three approaches the likelihood ratio tracks the LLM judge most closely — Spearman 0.195, against 0.190 for the Naive Bayes posterior, 0.137 for plain averaging and 0.082 for the quality-weighted variant. Since the likelihood ratio and Naive Bayes are the same mathematics, their agreement is itself evidence that both were implemented correctly.

Two contributions stand independently of that ranking. The first is a negative result with a mechanism attached: averaging cosine similarities across different encoder pairs does not measure coherence, because those similarities are not commensurable. One pair is larger on 500 of 500 prompts by a factor of 5.3, for reasons internal to the encoder rather than the data. Any metric of this family needs a per-pair baseline before the numbers can be combined. The second is diagnostic: one of our three quality features carries little signal because it duplicates another at a rank correlation of 0.902 — a design fault in the feature, not a shortage of data.

Four things follow, in order of priority.

- **1. Establish a per-pair baseline.** Shuffle prompts against outputs to obtain a distribution of similarities for triples known not to match, and read every raw similarity against that floor. This is the direct correction for the modality gap.
- **2. Replace the audio feature.** `Q_audio` remains the weakest of the three quality features at Spearman 0.090, and we now know why. A feature capturing prosody or expressiveness would carry information the text does not.

- **3. Justify the composite.** A single raw similarity beats our best composite. The burden is on any composite metric to demonstrate it earns its complexity.
- **4. Validate against humans.** Hand-score twenty to thirty triples for whether the three outputs match, and rank every candidate method against those labels rather than against a language model.

The conclusion we would draw for anyone building a similar system is not about which method won. It is that the sample size at which we first compared methods was too small to tell us anything reliable, and that we would have defended the wrong method had we stopped there.

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